

Experiment 1: SPICE Circuit Simulation and Equipment Usage

Malachy Crossan & Brandon Saavedra

EEE3307-C0013: Electronics I

Due Date: 2026-09-15

Last modified: 2026-09-15

1.A Objective

This lab is meant to familiarize ourselves with circuit simulation and lab equipment. We will be using LTSpice to simulate a simple circuit with a diode and resistor. Then we will build the circuit and measure the current through the diode and resistor using a digital multimeter (DMM) and an oscilloscope. We will compare the results of the simulation with the experimental results.

1.B Simulation

We first set up the circuit as seen in Figure 1. We then ran a DC sweep analysis to measure the current through the diode and resistor as a function of the voltage across the diode.

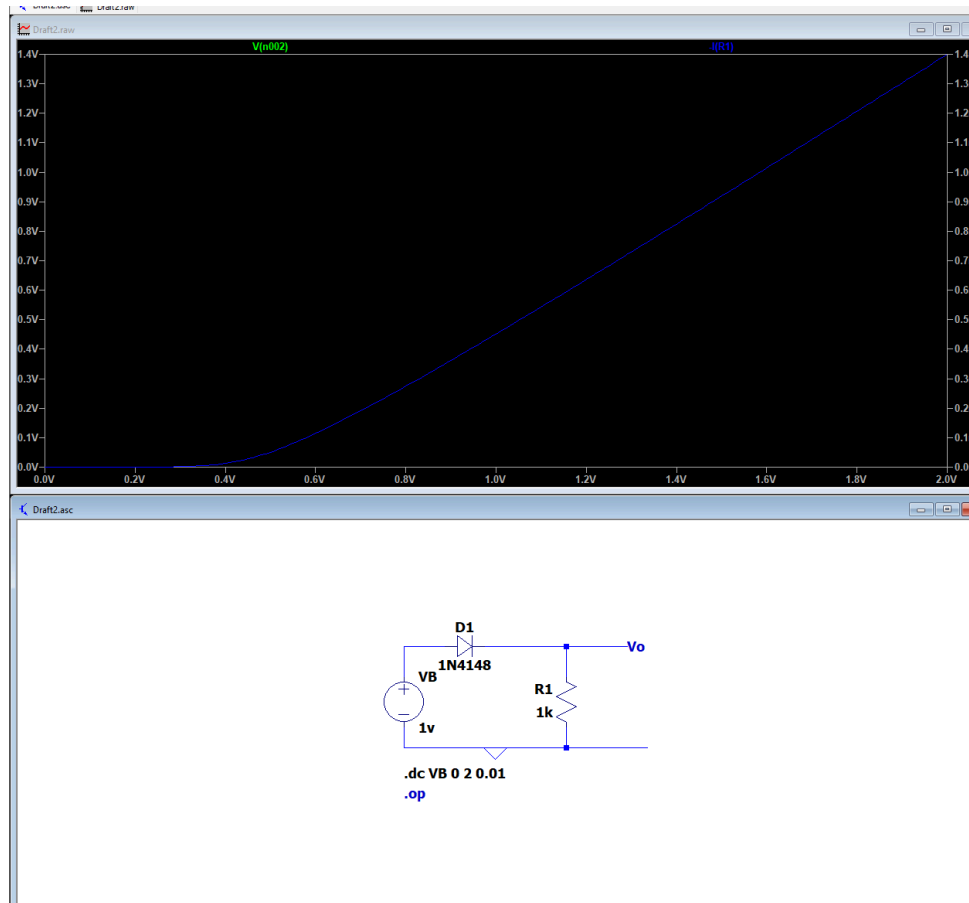


Figure 1: I_R over V_B (Simulation)

Date: 2026-09-14

Updating the voltage source to a 5 kHz sine wave with an amplitude of 2 Vpp, we then ran a transient analysis to measure the voltage across the resistor over time. The results can be seen in Figure 2.

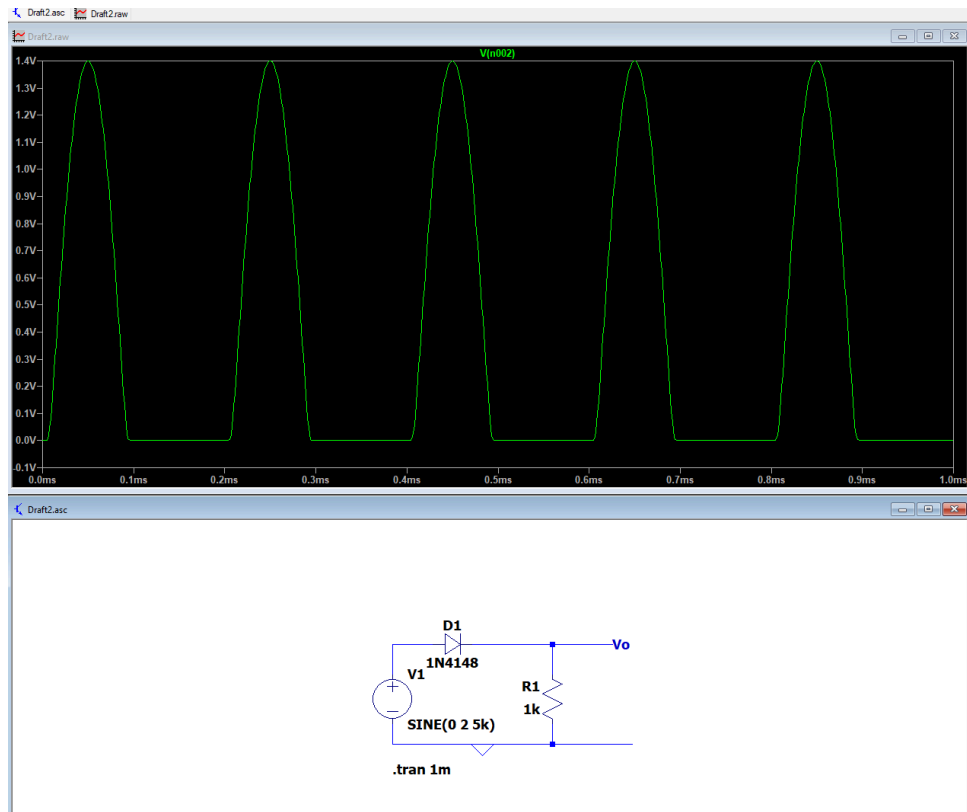


Figure 2: V_R over time (Simulation)

$$f = 5\text{kHz}$$

$$A = 2\text{Vpp}$$

Date: 2026-09-14

1.C Experiment

We then set up the circuit on a breadboard. The voltage source was represented by a variable power supply which was initially set to 1V. The resulting current in the circuit was measured using the DMM (Figure 3). The voltage source was then increased to 2V and the current was measured again (Figure 4).

Next, we connected the circuit to a function generator set to a 5 kHz sine wave with an amplitude of 2 Vpp. The voltage across the resistor was measured using an oscilloscope (Figure 5).

1.D Results

Comparing the results of the simulation and experimental values, we can see that at the two experimental points (Figure 3 and Figure 4) the current measured is very close to the simulated values.

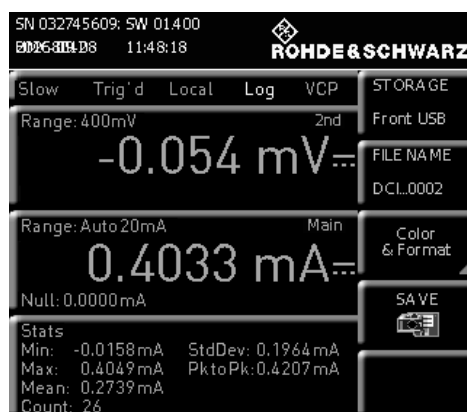


Figure 3: Current I of circuit $V_B = 1\text{V}$

Date: 2026-09-08

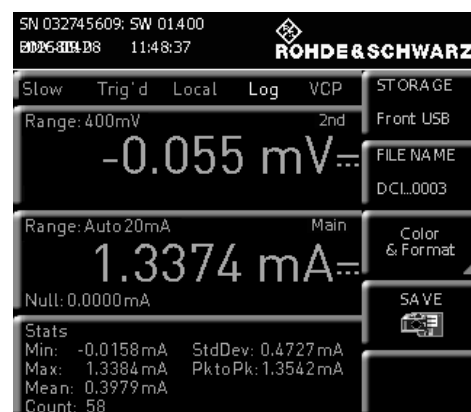


Figure 4: Current I of circuit $V_B = 2\text{V}$

Date: 2026-09-08

Also, the voltage across the resistor over time measured by the oscilloscope is very similar to the simulated results (Figure 5).



Figure 5: V_R over time

$$f = 5\text{kHz}$$

$$A = 2V_{pp}$$

Date: 2026-09-08

1.E Conclusion

This lab was successful in demonstrating the purpose of simulation and experimental validation. The results of the simulation and experimental measurements were very similar, indicating that the simulation was accurate. This lab also familiarized us with using LTSpice, a DMM, and an oscilloscope.